

Data-driven Finite-horizon Optimal Control for Linear Time-varying Discrete-time Systems

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Abstract—This paper presents a data-driven method to obtain an approximate solution of the finite-horizon optimal control problem for linear time-varying discrete-time systems. Firstly, a finite-horizon Policy Iteration method for linear time-varying discrete-time systems is proposed. Then, a data-driven off-policy Policy Iteration algorithm is derived to find approximate optimal controllers when the system dynamics is unknown. Under mild conditions, the proposed data-driven off-policy algorithm converges to the optimal solution. Finally, the effectiveness of the derived method is validated by a numerical example.

I. INTRODUCTION

Since dynamic programming (DP) was proposed by Bellman [2], significant progresses have been made on multi-stage decision problems, especially in optimal control of dynamical systems [4], [23]. However, the original DP is haunted by the “Curse of dimensionality” [2] and “Curse of modeling” [5]. Reinforcement learning [26], and adaptive/approximate dynamic programming [22] are promising to tackle these problems. Parametric function approximation techniques [25] are introduced to mitigate the burden of computation of value function caused by the first curse. For the second curse, data-driven iterative algorithms are proposed to find approximate optimal controllers by iteratively utilizing the data generated from the interactions between the controller and the plant under consideration.

Over the past decade, many papers have been devoted to dealing with the “Curse of modeling” for systems characterized by linear or nonlinear, differential or difference equations. Adaptive optimal controllers are found without the knowledge of system dynamics, by using Policy Iteration (PI) [13], [17], or Value Iteration (VI) [6], [3]. While most existing results focus on time-invariant systems and infinite-horizon optimal control problems, its counterparts, time-varying systems and finite-horizon optimal control problems, have been paid little attention. Different from the former case where the optimal controller is stationary, in the later case the optimal controller is nonstationary. This brings new challenges to the design of data-driven, model-free optimal controllers when the precise information of system dynamics is not available.

When the system dynamics are assumed to be known, finite-horizon approximate optimal controllers are derived for

linear or nonlinear time-invariant systems in [1], [8], [21]. Recently, several methods that do not require the knowledge of system dynamics have been proposed. By using extremum seeking techniques, approximate finite-horizon optimal open-loop control sequences are attained for linear time-varying discrete-time (LTVDT) systems in [12], [24]. Through dual-loop iteration, approximate optimal control is achieved for linear time-varying continuous-time (LTVCT) systems in [11]. A common feature of the methods in [11], [12] and [24] is that their methods use newly generated data in each iteration, which is not required in the present paper. In addition, the work [12], [24] only address the single-input systems while the work [11] does not provide explicit conditions for algorithmic convergence.

In this paper, the finite-horizon optimal control problem for linear time-varying discrete-time systems is considered. Firstly, finite-horizon PI method for LTVDT systems is presented, which can be seen as the counterpart to the infinite-horizon PI methods in [9], [10], [15], [19] and is parallel to the finite-horizon PI for LTVCT in [20]. Then based on this PI method, a data-driven off-policy PI algorithm is proposed to find approximate optimal controllers when the system dynamics is unknown. The proposed data-driven algorithm only needs to collect generated data for one time and reuse the same group of data repeatedly in each iteration to obtain the approximate optimal controller. In addition, multivariable control systems (with multiple inputs) are considered. The convergence of the data-driven algorithm to the optimal solution of the finite-horizon optimal control problem is guaranteed as long as certain mild rank conditions are satisfied. Finally, a numerical example demonstrates the effectiveness of the proposed method.

Notations. Throughout this paper, $\mathbb{R}$ denotes the set of real numbers, $\mathbb{Z}$ denotes the set of nonnegative integers. $k \in \mathbb{Z}$ denotes the discrete time instance. $X(k)$ is denoted as X_k for short and the time index is always the first subscript if it has multiple subscripts, e.g. X_{k,i_1,i_2} . $\otimes$ is the Kronecker product operator. For matrix $A \in \mathbb{R}^{n \times m}$, $\text{vec}(A) = [a_1^T, a_2^T, \dots, a_m^T]^T$, where a_i is the i th column of A . For symmetric matrix $B \in \mathbb{R}^{m \times m}$, $\text{vecs}(B) = [b_{11}, 2b_{12}, \dots, 2b_{1m}, b_{22}, 2b_{23}, \dots, 2b_{m-1,m}, b_{m,m}]^T \in \mathbb{R}^{\frac{1}{2}m(m+1)}$. For vector $v \in \mathbb{R}^n$, $\tilde{v} = [v_1^2, v_1v_2, \dots, v_1v_n, v_2^2, v_2v_3, \dots, v_{n-1}v_n, v_n^2]^T \in \mathbb{R}^{\frac{1}{2}n(n+1)}$. $\|\cdot\|$ is the Euclidean norm for vectors and $\|\cdot\|$ represents the induced matrix norm for matrices.

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II. PROBLEM FORMULATION AND PRELIMINARIES

Consider the following linear time-varying discrete-time system:

$$x_{k+1} = A_k x_k + B_k u_k \quad (1)$$

where $k \in \{k_0, k_0+1, \dots, N-1\}$ is the discrete time instant, $x_k \in \mathbb{R}^n$ is the system state, $u_k \in \mathbb{R}^m$ is the control input, $A_k \in \mathbb{R}^{n \times n}$ and $B_k \in \mathbb{R}^{n \times m}$ are time-dependent system matrices.

We are interested in finding a sequence of control inputs $\{u_k\}_{k=k_0}^{N-1}$, to minimize the following cost measure with penalty on the final state,

$$J(k_0, u) = x_N^T F x_N + \sum_{j=k_0}^{N-1} (x_j^T Q_j x_j + u_j^T R_j u_j) \quad (2)$$

where $Q_j \in \mathbb{R}^{n \times n}$, $R_j \in \mathbb{R}^{m \times m}$, $F \in \mathbb{R}^{n \times n}$ are symmetric weighting matrices, satisfying $Q_j \geq 0$, $R_j > 0$, and $F \geq 0$.

When the system matrices A_k and B_k are precisely known for all $k \in \{k_0, k_0+1, \dots, N-1\}$, this is the well-known finite-horizon linear quadratic regulator problem (see pages 148-151 in [4]). The optimal control input is given by,

$$u_k^* = -L_k^* x_k \quad (3)$$

where the gain matrices L_k^* are given by the equation

$$L_k^* = (R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k \quad (4)$$

and where the symmetric and positive semidefinite matrices P_k are given by solving the discrete-time Riccati equation,

$$P_k = Q_k + A_k^T P_{k+1} A_k - A_k^T P_{k+1} B_k (R_k + B_k^T P_{k+1} B_k)^{-1} B_k^T P_{k+1} A_k \quad (5)$$

with the condition $P_N = F$. Note that given a fixed matrix F , the matrix series $\{P_k\}_{k=k_0}^N$ are uniquely determined. Because when we compute P_k , all the matrices in the right-hand side of (5) have already been determined.

Remark 1: For the finite-horizon optimal control problem considered here, the existence of optimal control u^* is always guaranteed. This fact can be verified by a straightforward application of DP to the system (1) and cost function (2) (see pages 148-151 in [4]).

When A_k and B_k are unknown, however, Riccati equation (5) cannot be applied directly to obtain the optimal gain matrices L_k^* . Nor can existing infinite-horizon data-driven methods (see, e.g., [7], [14]) be used here, due to the difference between (5) and the algebraic Riccati equation (see page 151 in [4]) in infinite-horizon control problems. By exploiting the property of the cost function, a new method that finds the optimal control inputs without the knowledge of system matrices can be derived.

Now, assume that arbitrary gain matrices $\{L_k\}_{k=k_0}^{N-1}$ are adopted, i.e., $u_k = -L_k x_k$, $k \in \{k_0, k_0+1, \dots, N-1\}$. By applying them to the system (1), we have

$$x_{k+1} = A_{k,L} x_k, \quad (6)$$

where $A_{k,L} = A_k - B_k L_k$. For any $k_1, k_2 \in \{k_0, k_0+1, \dots, N\}$, $k_1 > k_2$, by the definition of state-transition matrix Φ_L associated with closed-loop system (6), we have

$$\begin{aligned} x_{k_1} &= \Phi_L(k_1, k_2) x_{k_2} \\ \Phi_L(k_1, k_2) &= A_{k_1-1,L} A_{k_1-2,L} \cdots A_{k_2,L} \\ \Phi_L(k_1, k_1) &= I \end{aligned} \quad (7)$$

Then we can write the cost at time instance k , $k \in \{k_0, k_0+1, \dots, N-1\}$, as

$$J(k, u_L) = x_k^T V_{k,L} x_k \quad (8)$$

where $V_{k,L}$ is the value matrix associated with the gain matrices $\{L_k\}_{k=k_0}^{N-1}$:

$$\begin{aligned} V_{k,L} &= \Phi_L^T(N, k) F \Phi_L(N, k) \\ &+ \sum_{j=k}^{N-1} [\Phi_L^T(j, k) (Q_j + L_j^T R_j L_j) \Phi_L(j, k)]. \end{aligned} \quad (9)$$

Note that matrix $V_{k,L}$ is symmetric and positive semidefinite, since $F \geq 0$, $Q_j \geq 0$, $R_j > 0$. Furthermore, $V_{N,L} = F$.

By using (7) with $k_1 = k+1$, $k_2 = k$ and (9), the following Lyapunov equation can be obtained

$$V_{k,L} = A_{k,L}^T V_{k+1,L} A_{k,L} + Q_k + L_k^T R_k L_k. \quad (10)$$

III. MAIN RESULTS

In this section, firstly the finite-horizon PI method for linear time-varying discrete-time systems is presented. Then, based on this PI method, we propose a data-driven adaptive dynamic programming algorithm to find the approximate optimal controllers without the knowledge of system matrices.

A. Finite-horizon Policy Iteration for linear time-varying discrete-time systems

Theorem 1: For system (1), consider the finite-horizon linear quadratic regulator problem with respect to the cost function (2).

- 1) Choose arbitrary initial gain matrices $\{L_{k,0}\}_{k=k_0}^{N-1}$, and let $i = 0$.
- 2) (Policy Evaluation) Solve for $\{V_{k,i}\}_{k=k_0}^{N-1}$, by using the following Lyapunov equations with $V_{N,i} = F$,

$$V_{k,i} = A_{k,i}^T V_{k+1,i} A_{k,i} + Q_k + L_{k,i}^T R_k L_{k,i} \quad (11)$$

where $A_{k,i} = A_k - B_k L_{k,i}$, $V_{k,i}$ is the value matrix defined in (9) with respect to $\{L_{k,i}\}_{k=k_0}^{N-1}$.

- 3) (Policy Improvement) Solve for $\{L_{k,i+1}\}_{k=k_0}^{N-1}$, by using the following equations,

$$L_{k,i+1} = (R_k + B_k^T V_{k+1,i} B_k)^{-1} B_k^T V_{k+1,i} A_k \quad (12)$$

- 4) Let $i = i + 1$, and go to Step 2).

Then for all $k \in \{k_0, k_0+1, \dots, N\}$, it holds:

- (i) $P_k \leq V_{k,i+1} \leq V_{k,i}$, $\forall i \in \mathbb{Z}$.
- (ii) $\lim_{i \rightarrow \infty} V_{k,i} = P_k$.

Proof: (i) Note that

$$\begin{aligned} A_{k,i} &= A_{k,i+1} - B_k (L_{k,i} - L_{k,i+1}) \\ L_{k,i} &= (L_{k,i} - L_{k,i+1}) + L_{k,i+1} \end{aligned}$$

Substituting the above equations into (11) gives

$$\begin{aligned} V_{k,i} &= V_{k,i+1} + A_{k,i+1}^T (V_{k+1,i} - V_{k+1,i+1}) A_{k,i+1} \\ &\quad + (L_{k,i} - L_{k,i+1})^T (R_k + B_k^T V_{k+1,i} B_k) (L_{k,i} - L_{k,i+1}) \\ &\quad + (L_{k,i} - L_{k,i+1})^T (R_k L_{k,i+1} - B_k^T V_{k+1,i} A_{k,i+1}) \\ &\quad + (R_k L_{k,i+1} - B_k^T V_{k+1,i} A_{k,i+1})^T (L_{k,i} - L_{k,i+1}) \end{aligned}$$

Note that the last two terms in the above equation vanish as a result of (12). Then we have

$$\begin{aligned} V_{k,i} - V_{k,i+1} &= A_{k,i+1}^T (V_{k+1,i} - V_{k+1,i+1}) A_{k,i+1} \\ &\quad + (L_{k,i} - L_{k,i+1})^T (R_k + B_k^T V_{k+1,i} B_k) (L_{k,i} - L_{k,i+1}) \end{aligned} \quad (13)$$

Note that $(R_k + B_k^T V_{k+1,i} B_k)$ is always positive definite. Since $V_{N,i} = V_{N,i+1} = F$, by induction, $V_{k,i} \geq V_{k,i+1}$ for all i and k .

Since P_k is the value matrix associated with the optimal control gain matrices $\{L_k^*\}_{k=k_0}^{N-1}$, $x_k^T P_k x_k \leq x_k^T V_{k,i+1} x_k$, $\forall x_k \in \mathbb{R}^n$. Hence, $P_k \leq V_{k,i+1}$.

(ii) By (i), $V_{k,i} \geq V_{k,i+1} \geq P_k$. This implies sequence $\{V_{k,i}\}_{i=0}^\infty$ is nonincreasing and bounded from below. By a theorem on the convergence of a monotone sequence of self-adjoint operators (See pages 189-190 in [18]), $\lim_{i \rightarrow \infty} V_{k,i}$ exists. Letting $i \rightarrow \infty$ in (11) and (12), we have by continuity

$$\begin{aligned} V_{k,\infty} &= A_k^T V_{k+1,\infty} A_k + Q_k - L_{k,\infty}^T B_k^T V_{k+1,\infty} A_k \\ &\quad - A_k^T V_{k+1,\infty} B_k L_{k,\infty} \\ &\quad + L_{k,\infty}^T (B_k^T V_{k+1,\infty} B_k + R_k) L_{k,\infty} \\ L_{k,\infty} &= (R_k + B_k^T V_{k+1,\infty} B_k)^{-1} B_k^T V_{k+1,\infty} A_k \end{aligned}$$

Eliminating $L_{k,\infty}$ from the above equations yields

$$\begin{aligned} V_{k,\infty} &= Q_k + A_k^T V_{k+1,\infty} A_k \\ &\quad - A_k^T V_{k+1,\infty} B_k (R_k + B_k^T V_{k+1,\infty} B_k)^{-1} B_k^T V_{k+1,\infty} A_k. \end{aligned}$$

This is exactly the Riccati equation. Due to $P_N = V_{N,\infty} = F$ and the uniqueness of solution of Riccati equation, $P_k = V_{k,\infty}$ follows. ■

B. Data-driven off-policy Policy Iteration

When the system matrices are not known, the procedure in Theorem 1 can be implemented by using input/state data collected from the trajectories of the control systems.

Suppose that a series of control inputs $\{u_{k,0}\}_{k=k_0}^{N-1}$ is applied to the system to generate data, and we are in the i th stage of the procedure in Theorem 1. Then (1) can be rewritten as

$$x_{k+1} = A_{k,i} x_k + B_k (L_{k,i} x_k + u_{k,0})$$

Next, we have

$$\begin{aligned} &x_{k+1}^T V_{k+1,i} x_{k+1} - x_k^T V_{k,i} x_k = \\ &x_k^T (A_{k,i}^T V_{k+1,i} A_{k,i} - V_{k,i}) x_k \\ &\quad + 2(L_{k,i} x_k + u_{k,0})^T B_k^T V_{k+1,i} A_{k,i} x_k \\ &\quad + (L_{k,i} x_k + u_{k,0})^T B_k^T V_{k+1,i} B_k (L_{k,i} x_k + u_{k,0}) \end{aligned} \quad (14)$$

Substituting (11) into equation (14) and rearranging the terms, we obtain

$$\begin{aligned} &x_k^T (Q_k + L_{k,i}^T R_k L_{k,i}) x_k = \\ &x_k^T V_{k,i} x_k - x_{k+1}^T V_{k+1,i} x_{k+1} \\ &\quad + 2(L_{k,i} x_k + u_{k,0})^T B_k^T V_{k+1,i} A_{k,i} x_k \\ &\quad + (u_{k,0} + L_{k,i} x_k)^T B_k^T V_{k+1,i} B_k (u_{k,0} - L_{k,i} x_k) \end{aligned} \quad (15)$$

By using the properties of Kronecker product, equation (15) can be rewritten as

$$\psi_{k,i} = \phi_{k,i} \begin{bmatrix} \text{vecs}(V_{k,i}) \\ \text{vec}(B_k^T V_{k+1,i} A_k) \\ \text{vecs}(B_k^T V_{k+1,i} B_k) \\ \text{vecs}(V_{k+1,i}) \end{bmatrix} \quad (16)$$

where

$$\begin{aligned} \psi_{k,i} &= x_k^T (Q_k + L_{k,i}^T R_k L_{k,i}) x_k \\ \phi_{k,i} &= [\tilde{x}_k^T, \delta_{k,i}, \tilde{u}_{k,0}^T - \widetilde{L_{k,i} x_k}^T, -\tilde{x}_{k+1}^T] \\ \delta_{k,i} &= 2[(x_k^T \otimes x_k^T)(I_n \otimes L_{k,i}^T) + (x_k^T \otimes u_{k,0}^T)] \end{aligned}$$

Note that $\psi_{k,i}$ and $\phi_{k,i}$ are all known data matrices.

Since (16) is only a scalar equation, involving more than one unknown parameters to be determined, we need to collect more data to form a system of linear equations. To this end, suppose in total l groups of different control sequences $\{u_{k,0,j}\}_{k=k_0}^{N-1}$, $j = 1, 2, \dots, l$ are applied to the system and corresponding data are recorded. Each group of control sequences has the following form,

$$u_{k,0,j} = -L_{k,0} x_j + w_{k,j} \quad (17)$$

where $w_{k,j} \in \mathbb{R}^m$ is the exploration noise used to achieve sufficient excitation of the system. By defining the following data matrices,

$$\begin{aligned} \Gamma_{\tilde{x}_k} &= [\tilde{x}_{k,1}, \tilde{x}_{k,2}, \dots, \tilde{x}_{k,l}]^T \\ \Gamma_{xxk} &= [x_{k,1} \otimes x_{k,1}, x_{k,2} \otimes x_{k,2}, \dots, x_{k,l} \otimes x_{k,l}]^T \\ \Gamma_{xu_{k,0}} &= [x_{k,1} \otimes u_{k,0,1}, x_{k,2} \otimes u_{k,0,2}, \dots, x_{k,l} \otimes u_{k,0,l}]^T \\ \Gamma_{\tilde{u}_{k,0}} &= [\tilde{u}_{k,0,1}, \tilde{u}_{k,0,2}, \dots, \tilde{u}_{k,0,l}]^T \\ \Gamma_{\widetilde{L_{k,i} x_k}} &= [\widetilde{L_{k,i} x_{k,1}}, \widetilde{L_{k,i} x_{k,2}}, \dots, \widetilde{L_{k,i} x_{k,l}}]^T \\ \Delta_{k,i} &= 2(\Gamma_{xxk} (I_n \otimes L_{k,i}^T) + \Gamma_{xu_{k,0}}) \end{aligned}$$

we obtain the matrix equation,

$$\Psi_i = \Phi_i \Theta_i \quad (18)$$

where

$$\Theta_i = \begin{bmatrix} \theta_{k_0,i} \\ \theta_{k_0+1,i} \\ \vdots \\ \theta_{N-1,i} \end{bmatrix}, \theta_{k,i} = \begin{bmatrix} \text{vecs}(V_{k,i}) \\ \text{vec}(B_k^T V_{k+1,i} A_k) \\ \text{vecs}(B_k^T V_{k+1,i} B_k) \end{bmatrix}$$

and Ψ_i, Φ_i are given in (19), (20) below.

If Φ_i is a full-column rank matrix, (18) can be uniquely solved, i.e.

$$\Theta_i = (\Phi_i^T \Phi_i)^{-1} \Phi_i^T \Psi_i \quad (21)$$

$$\Phi_i = \begin{bmatrix} \Gamma_{\tilde{x}_{k_0}} & \Delta_{k_0,i} & (\Gamma_{\tilde{u}_{k_0,0}} - \Gamma_{\widetilde{L_{k_0,i}x_{k_0}}}) & -\Gamma_{\tilde{x}_{k_0+1}} & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \Gamma_{\tilde{x}_{k_0+1}} & \Delta_{k_0+1,i} & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & \Gamma_{\tilde{x}_{N-1}} & \Delta_{N-1,i} & (\Gamma_{\tilde{u}_{N-1,0}} - \Gamma_{\widetilde{L_{N-1,i}x_{N-1}}}) \end{bmatrix} \quad (19)$$

$$\Psi_i = [\Gamma_{\tilde{x}_{k_0}} \text{vecs}(Q_{k_0} + L_{k_0,i}^T R_{k_0} L_{k_0,i}), \Gamma_{\tilde{x}_{k_0+1}} \text{vecs}(Q_{k_0+1} + L_{k_0+1,i}^T R_{k_0+1} L_{k_0+1,i}), \dots, \Gamma_{\tilde{x}_{N-1}} \text{vecs}(Q_{N-1} + L_{N-1,i}^T R_{N-1} L_{N-1,i}) + \Gamma_{\tilde{x}_N} \text{vecs}(F)]^T \quad (20)$$

As a result, the gain matrix $L_{k,i+1}$ can be obtained by using equation (12).

Now we are ready to present the data-driven off-policy Policy Iteration algorithm.

Algorithm 1: Data-driven off-policy Policy Iteration

Choose arbitrary $\{L_{k,0}\}_{k=k_0}^{N-1}$, threshold $\epsilon > 0$. $i \leftarrow 0$.
 Run system (1) l times. In j th run, use $\{u_{k,0,j}\}_{k=k_0}^{N-1}$ from (17) as control inputs.
 Collect the generated data.
repeat
 Compute Ψ_i by (19);
 Compute Φ_i by (20);
 $\Theta_i \leftarrow (\Phi_i^T \Phi_i)^{-1} \Phi_i^T \Psi_i$;
 $k \leftarrow k_0$;
 while $k < N$ **do**
 $L_{k,i+1} \leftarrow (R_k + B_k^T V_{k+1,i} B_k)^{-1} B_k^T V_{k+1,i} A_k$;
 $k \leftarrow k + 1$;
 $i \leftarrow i + 1$
until $\max_k \|V_{k,i} - V_{k,i-1}\| < \epsilon$;
 Use $u_k = -L_{k,i} x_k$ as the approximate optimal control.

Next, the condition that guarantees the convergence of Algorithm 1 is presented.

Theorem 2: If there exists an integer $l_0 > 0$, such that, for all $l > l_0$ and $k \in \{k_0, k_0 + 1, \dots, N - 1\}$,

$$\text{rank}([\Gamma_{\tilde{x}_k}, \Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}]) = \frac{m(m+1)}{2} + mn + \frac{n(n+1)}{2} \quad (22)$$

then,

- (i) Φ_i has full column rank for all $i \in \mathbb{Z}$.
- (ii) the sequences $\{V_{k,i}\}_{i=0}^\infty$ and $\{L_{k,i}\}_{i=0}^\infty$ obtained by iteratively solving equation (18) and equation (12) converge to the optimal values P_k and L_k^* , respectively.

Proof: For convenience, define variables

$$\begin{aligned} \Xi_v &= [X_v^T, Y_v^T, Z_v^T]^T \\ \Phi_{k,i} &= [\Gamma_{\tilde{x}_k}, \Delta_{k,i}, \Gamma_{\tilde{u}_{k,0}} - \Gamma_{\widetilde{L_{k,i}x_k}}] \\ \Psi_{k,i} &= \Gamma_{\tilde{x}_k} \text{vecs}(Q_k + L_{k,i}^T R_k L_{k,i}) + \Gamma_{\tilde{x}_{k+1}} \text{vecs}(W_m) \end{aligned}$$

where $X_v = \text{vecs}(X_m)$, $Y_v = \text{vec}(Y_m)$, $Z_v = \text{vecs}(Z_m)$, $X_m, W_m \in \mathbb{R}^{n \times n}$ and $Z_m \in \mathbb{R}^{m \times m}$ are symmetric matrices,

$Y_m \in \mathbb{R}^{m \times n}$, $k \in \{k_0, k_0 + 1, \dots, N - 1\}$. It is easy to check that,

$$\Phi_{k,i} \theta_{k,i} = \Psi_{k,i} \quad (23)$$

(i) Obviously, Φ_i has full column rank if and only if $\Phi_{k,i}$ has full column rank for all $k \in \{k_0, k_0 + 1, \dots, N - 1\}$. This is equivalent to show that linear equation

$$\Phi_{k,i} \Xi_v = 0$$

has unique solution $\Xi_v = 0$. We shall show that it is indeed the case.

According to the definition of $\Phi_{k,i}$ and the equation (15), we have

$$\Phi_{k,i} \Xi_v = [\Gamma_{\tilde{x}_k}, 2\Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}] \begin{bmatrix} \text{vecs}(\Lambda_1) \\ \text{vec}(\Lambda_2) \\ \text{vecs}(\Lambda_3) \end{bmatrix} = 0 \quad (24)$$

where

$$\begin{aligned} \Lambda_1 &= X_m + L_{k,i}^T Y_m + Y_m^T L_{k,i} - L_{k,i}^T Z_m L_{k,i} \\ \Lambda_2 &= Y_m \\ \Lambda_3 &= Z_m \end{aligned}$$

From (22), we know that $[\Gamma_{\tilde{x}_k}, 2\Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}]$ has full column rank. This means the only solution to (24) is $\Lambda_1 = 0$, $\Lambda_2 = 0$, $\Lambda_3 = 0$. This is true if and only if $\Xi_v = 0$. Thus $\Phi_{k,i}$ always has full column rank. Therefore Φ_i has full column ranks for all $i \in \mathbb{Z}$.

(ii) Suppose Ξ_v satisfies

$$\Phi_{k,i} \Xi_v - \Psi_{k,i} = 0$$

By definitions of $\Phi_{k,i}$ and $\Psi_{k,i}$, this is equivalent to

$$[\Gamma_{\tilde{x}_k}, 2\Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}] \begin{bmatrix} \text{vecs}(\Omega_1) \\ \text{vec}(\Omega_2) \\ \text{vecs}(\Omega_3) \end{bmatrix} = 0 \quad (25)$$

where

$$\begin{aligned} \Omega_1 &= X_m - Q_k - L_{k,i}^T R_k L_{k,i} + L_{k,i}^T Y_m + Y_m^T L_{k,i} \\ &\quad - L_{k,i}^T Z_m L_{k,i} - A_k^T W_m A_k \\ \Omega_2 &= Y_m - B_k^T W_m A_k \\ \Omega_3 &= Z_m - B_k^T W_m B_k \end{aligned}$$

Again, $[\Gamma_{\tilde{x}_k}, 2\Gamma_{xu_{k,0}}, \Gamma_{\tilde{u}_{k,0}}]$ has full column rank. This means the only solution to (25) is $\Omega_1 = 0$, $\Omega_2 = 0$, $\Omega_3 = 0$.

Substituting $Y_m = B_k^T V_{k+1,i} A_k$, $Z_m = B_k^T V_{k+1,i} B_k$ into $\Omega_1 = 0$, we get

$$X_m = Q_k + L_{k,i}^T R_k L_{k,i} + A_{k,i}^T W_m A_{k,i} \quad (26)$$

which is exactly the Lyapunov equation (11). Since $V_{N,i} = F$ for all $i \in \mathbb{Z}$, Policy Iteration by (18) and (12) is equivalent to (11) and (12). By Theorem 1, the convergence is proved. ■

Remark 2: Different from the infinite-horizon PI algorithms [15], [19], [14], [16], where the initial gain matrices must be stabilizing to guarantee the convergence of the algorithm, here the initial gain matrices can be arbitrary in theory. However, in practice, utilization of stabilizing initial gain matrices (if exist) will prevent the state from becoming too large.

Remark 3: In Algorithm 1, there is no need to identify system model (1) explicitly, so the proposed method can be regarded as a direct adaptive control method. For indirect adaptive control, system identification is applied to identify the parameters of the system model and then to compute the desired adaptive controller. When the dimension of control inputs m is less than the dimension of system state n , the number of unknowns to be determined in Algorithm 1 will be fewer than that in indirect adaptive control.

IV. NUMERICAL EXAMPLE

In this section, the effectiveness of the proposed algorithm is demonstrated by using a time-varying system. As long as rich enough data are collected in the considered time interval, the approximate optimal controller can be derived without the precise knowledge of system matrices.

Consider the linear time-varying discrete-time system (1) described by the following matrices,

$$A_k = \begin{bmatrix} 1 & 0.0025k \\ -0.1 \cos(0.3k) & 1 + (0.05)^{\frac{3}{2}} \sin(0.5k)\sqrt{k} \end{bmatrix},$$

$$B_k = \begin{bmatrix} 1 \\ \frac{(0.1k+2)}{(0.1k+3)} \end{bmatrix} 0.05.$$

Let $k_0 = 0$, $N = 120$. The weighting matrices are $Q_k = (0.04k + 2)I_2$, $R_k = 5 - 0.02k$, $F = 50I_2$. Choose the initial gain matrices $L_{k,0} = [0, 0]^T$. To make the data collected satisfying condition (22), in the simulation, the elements of initial state are assumed to be independently and uniformly distributed over $[-1, 1]$, and the exploration noises are chosen as,

$$w_{k,j} = 2 \left(\sum_{s=1}^{500} \sin(\sigma_{s,j} \cdot k) \right)$$

where in j th trial, each $\sigma_{s,j}$ is independently drawn from the uniform distribution over $[-500, 500]$.

Set $\epsilon = 0.01$ in Algorithm 1. After 8 iterations, the algorithm terminates and an approximate optimal solution is obtained. The differences between $V_{k,i}$ and P_k from iteration 1 to iteration 4 and from iteration 4 to iteration 8 are shown in Figures 1 and 2, respectively. One can see that the value matrices given by Algorithm 1 converge monotonically to

the optimal value matrices for all k , as proved in Theorem 1. Both the resulting approximate optimal controller and the initial controller are applied to the system with initial condition $x_0 = [-2, 1]^T$. The state trajectories comparison and the control input are presented in Figures 3 and 4, respectively.

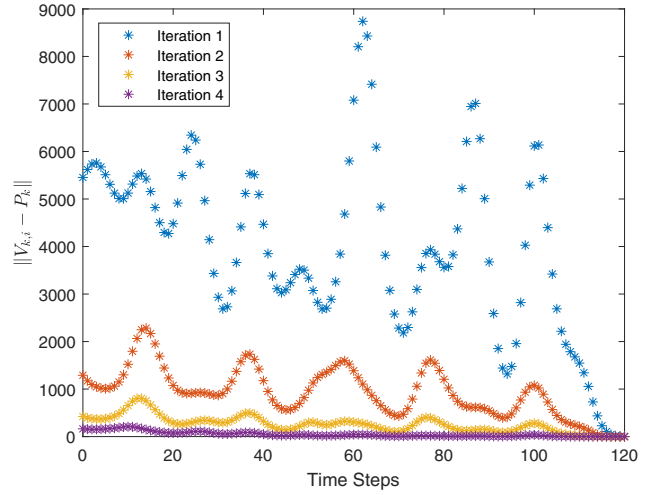


Fig. 1. The convergence of $V_{k,i}$ from iteration 1 to iteration 4.

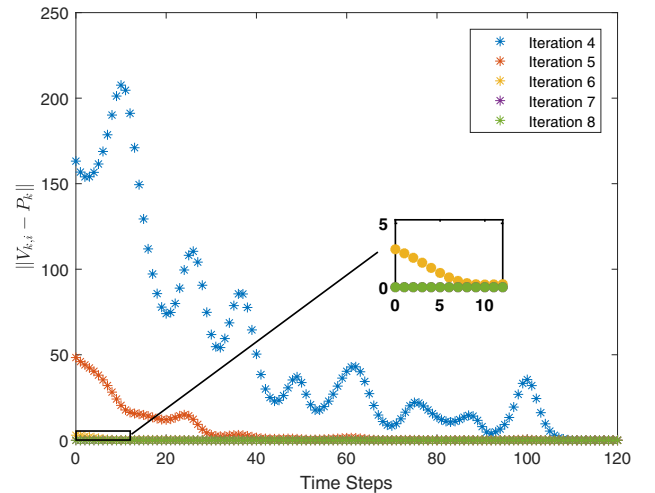


Fig. 2. The convergence of $V_{k,i}$ from iteration 4 to iteration 8.

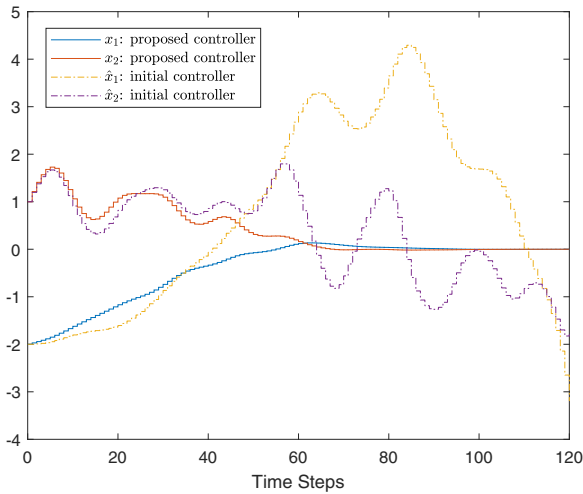


Fig. 3. The state trajectories comparison.

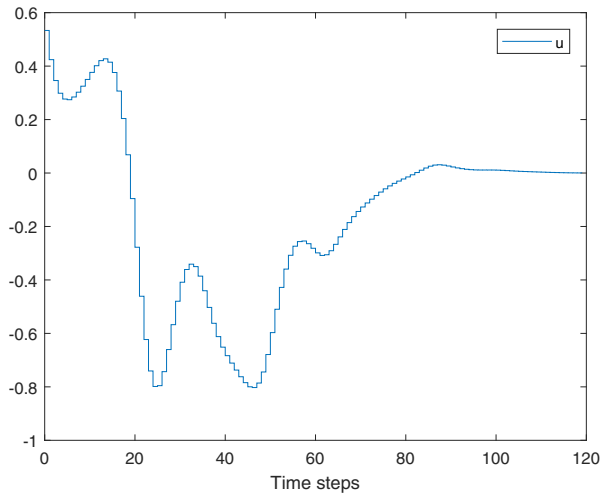


Fig. 4. Plot of the obtained approximated optimal controller.

V. CONCLUSIONS

This paper presents a method dealing with the “Curse of modeling” for finite-horizon optimal control of linear time-varying discrete-time systems with completely unknown dynamics. A novel finite-horizon Policy Iteration method for linear time-varying discrete-time systems is proposed. An algorithm that does not require the exact knowledge of system dynamics is presented for the design of adaptive optimal controllers based on the finite-horizon Policy Iteration method. As long as rich enough data are collected such that certain rank conditions are satisfied, the proposed algorithm is guaranteed to converge to the optimal solution. Numerical simulations demonstrate the feasibility of the proposed method.

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